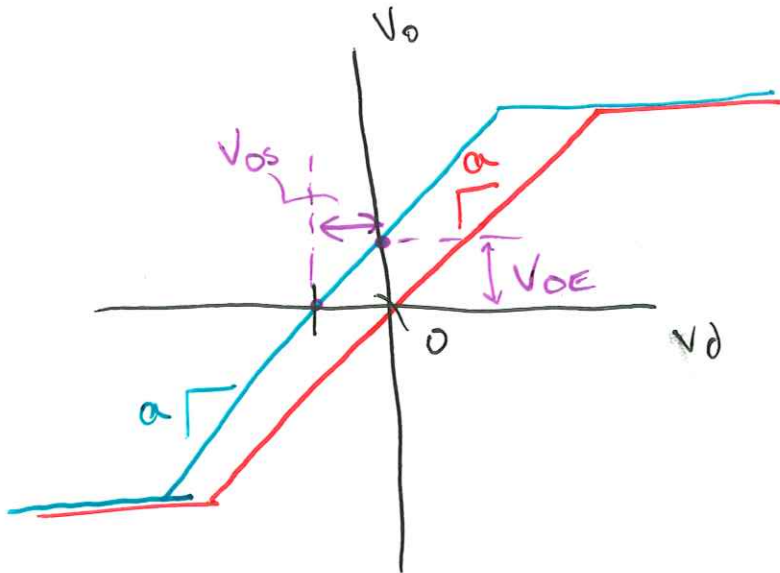


Voltage Offsets

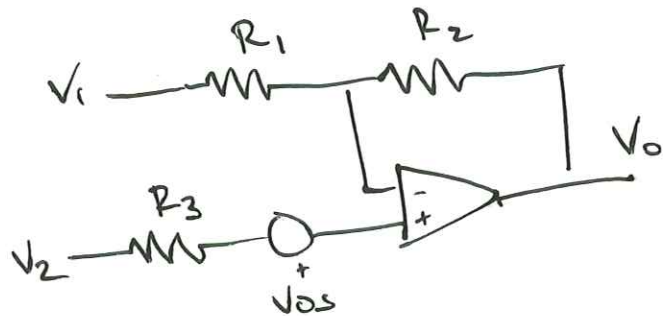


V_{oe} = Output error (due to input offset voltage)

V_{os} = Input offset voltage.

→ $\pm$ error

Modelling: ($i_n = i_p = 0$)



$$V_o = -\frac{R_2}{R_1} V_1 + \left(1 + \frac{R_2}{R_1}\right) V_2 + \left(1 + \frac{R_2}{R_1}\right) V_{os}$$

- V_{os} :
- Manufacturing Defects → Range Stated in Datasheet
 - Temperature Dependent → Datasheet
 - typically in μV range up to mV range

V_{os} commonly given in Datasheet as expected range at 25°C (or 20°C)

for example: 741 $\rightarrow V_{os}(25^{\circ}\text{C}) = \pm 0.8\text{mV}$

V_{os} temperature dependance typically stated as $\text{mV}/^{\circ}\text{C}$ change.

for example: 741 $\rightarrow TC(V_{os}) = \pm 15\text{mV}/^{\circ}\text{C} \quad -55^{\circ}\text{C} \leq T \leq 125^{\circ}\text{C}$

$$V_{os}(T) = V_{os}(25^{\circ}\text{C}) + TC(V_{os}) \Delta T$$

for Ex 741 @ 50°C

$$V_{os}(50^{\circ}\text{C}) = 8\text{mV} + 15\text{mV}/^{\circ}\text{C}(50^{\circ}\text{C} - 25^{\circ}\text{C}) = 1.175\text{mV}$$

CMRR of input stage within OpAmp
(it is a Diff. Amp $V \rightarrow I$!)

$$V_o = a(V_p - V_n) \quad \text{let } V_p = V_{cm,d} + V_{dm,d}/2 \quad V_{dm,d} = V_d$$

$$V_n = V_{cm,d} - V_{dm,d}/2$$

$$V_o = a_{dm,d} V_{dm,d} + a_{cm,d} V_{cm,d}$$

Assume that $a_{dm,d} \approx a$

$$V_o = \underbrace{a V_{dm,d}}_{\text{Desired output}} + \underbrace{a_{cm,d} \left(\frac{V_p + V_n}{2} \right)}_{\text{common mode output.}}$$

$$\therefore \text{CMRR} = \left| \frac{a}{a_{cm}} \right| \rightarrow a_{cm} = \pm \frac{a}{\text{CMRR}} \quad \leftarrow \text{CM gain at output.}$$

$$V_o = a V_d \pm \frac{a}{\text{CMRR}} V_{cm}$$

$$V_o = a \left(V_d \pm \frac{V_{cm}}{\text{CMRR}} \right)$$

desired $\uparrow$

error $\rightarrow$ model as input voltage offset!

$$V_{os}(\text{CMRR}) = \pm \frac{V_{cm}}{\text{CMRR}}$$

Other Error Sources (V_{OE}) that can be lumped into input offset voltage.

Power Supply Changes

ex. Supply drops from $\pm 15V$ to $\pm 10V$

$$|\Delta V_{os}| \propto \Delta V_{supply}$$

$$|\Delta V_{os}| = \Delta V_{supply} / PSRR$$

ΔV_{supply}
→ use unipolar values

$$\text{Ex: } 741 \quad \pm 15V - \pm 10V \rightarrow \Delta V_{supply} = 5V$$

$$|\Delta V_{os}| = 5V / 15 \mu V/V = 333 \mu V$$

741 states PSRR inverted

$$|\Delta V_{os}| = 5V / (1/15 \mu V/V) = 75 \mu V$$

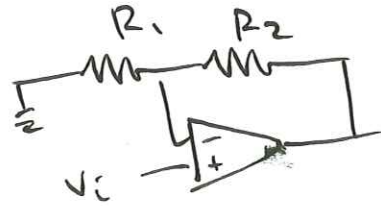
PSRR Examples:

$$741: 15 \mu V/V$$

$$ADA4807: 107 dB \rightarrow 223 \mu V/V$$

← Incorrect!

Gain Error due to finite a



$$A = A_{ideal} / (1 + 1/L)$$

$$V_o = A_{ideal} V_i - \frac{V_o}{L}$$

$$L = a_2 b = a_2 / A_{ideal}$$

$$V_o = A_{ideal} \left(V_i - \frac{V_o}{a_2} \right)$$

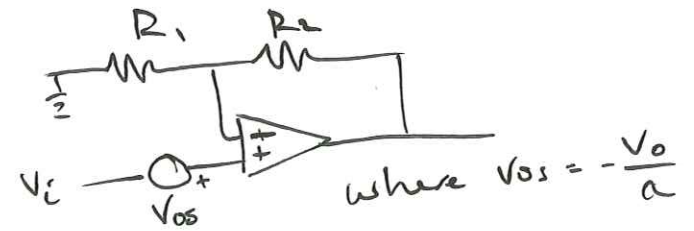
desired

offset input voltage.

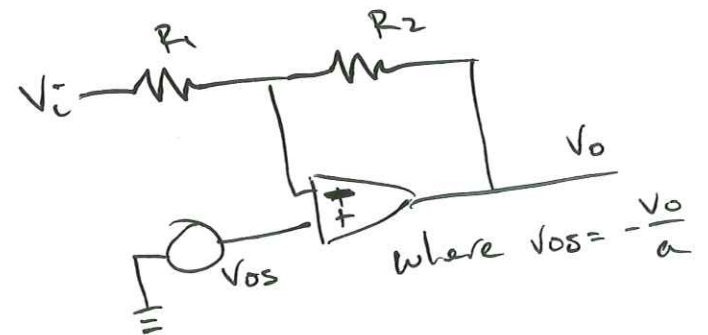
V_{os} always on + input terminal

$$\therefore a_2 = a$$

$$V_o = A_{ideal} \left(V_i - \frac{V_o}{a} \right) \rightarrow \text{mode alternatively}$$



$$V_o = \left(1 + \frac{R_2}{R_1} \right) V_i + \left(1 + \frac{R_2}{R_1} \right) V_{os}$$



$$V_o = -\frac{R_2}{R_1} V_i + \left(1 + \frac{R_2}{R_1} \right) V_{os}$$

Lumping all together.

(worst case scenario)

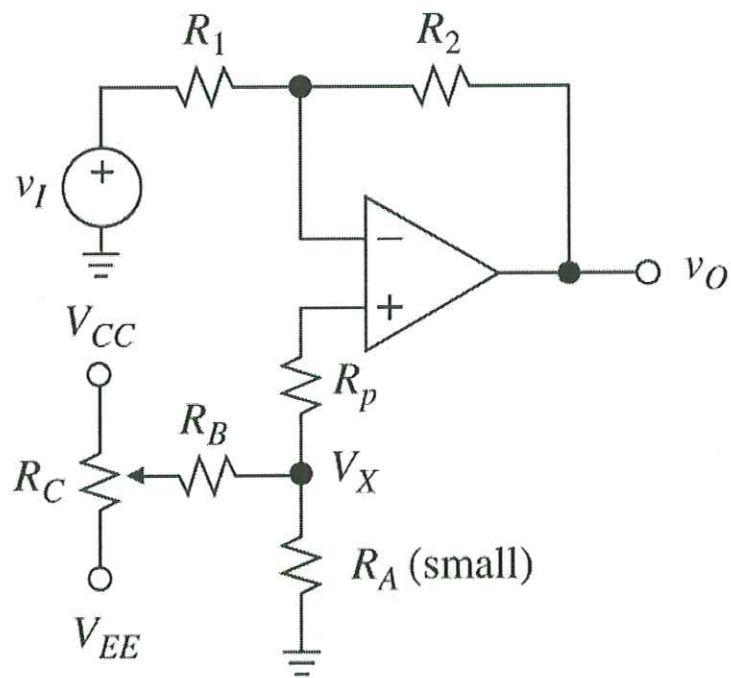
$$V_{os} = V_{os}(25^{\circ}\text{C}) + TC(V_{os}) \Delta T + \frac{\Delta V_{\text{supply}}}{PSRR} + \frac{V_{cm}}{CMRR} + \frac{V_{omax}}{a}$$

V_{omax}

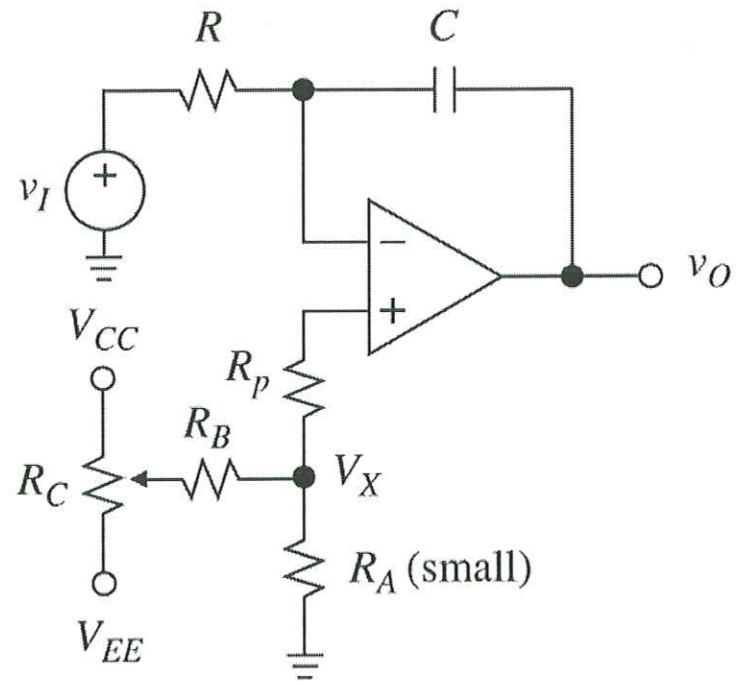
→ Maximum expected output voltage.

ways to eliminate (or reduce...) V_{os} :

Add input offset voltage to
cancel out amp V_{os}



(a)



(b)

Figure 5.20: External offset error nulling for (a) the inverting amplifier and (b) the integrator.

Input offset nulling via external potentiometer.

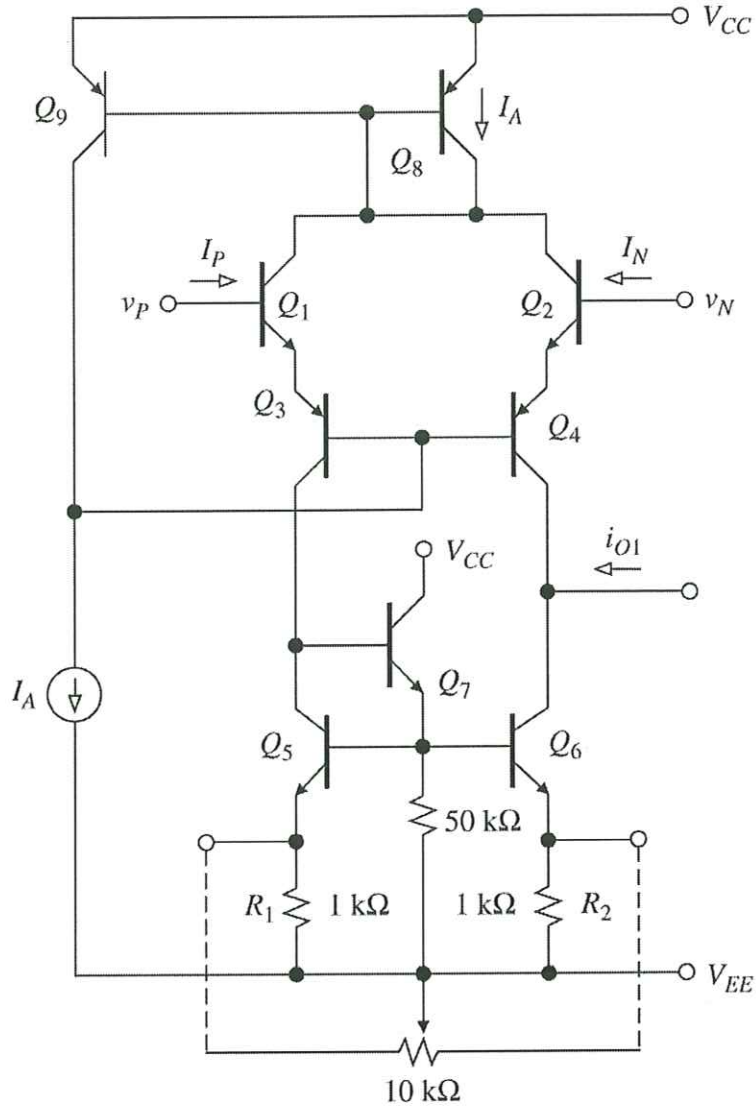
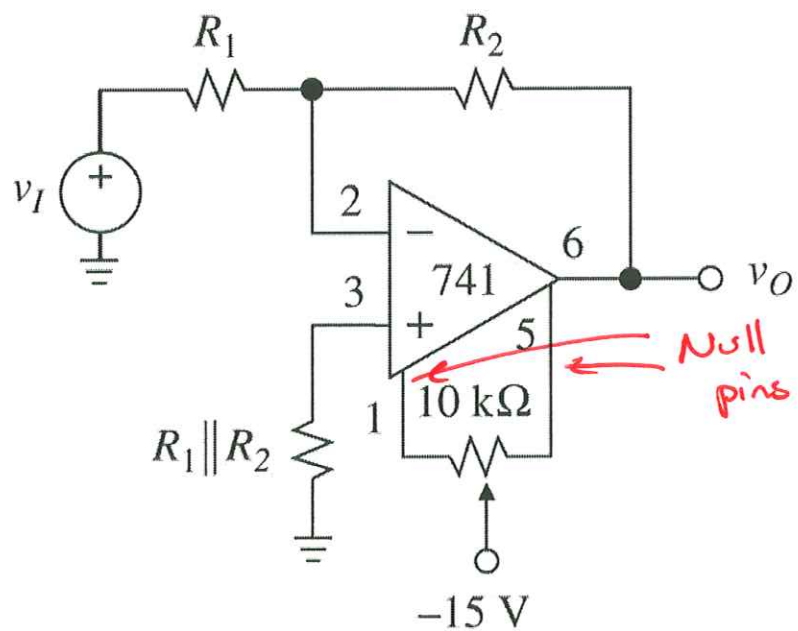
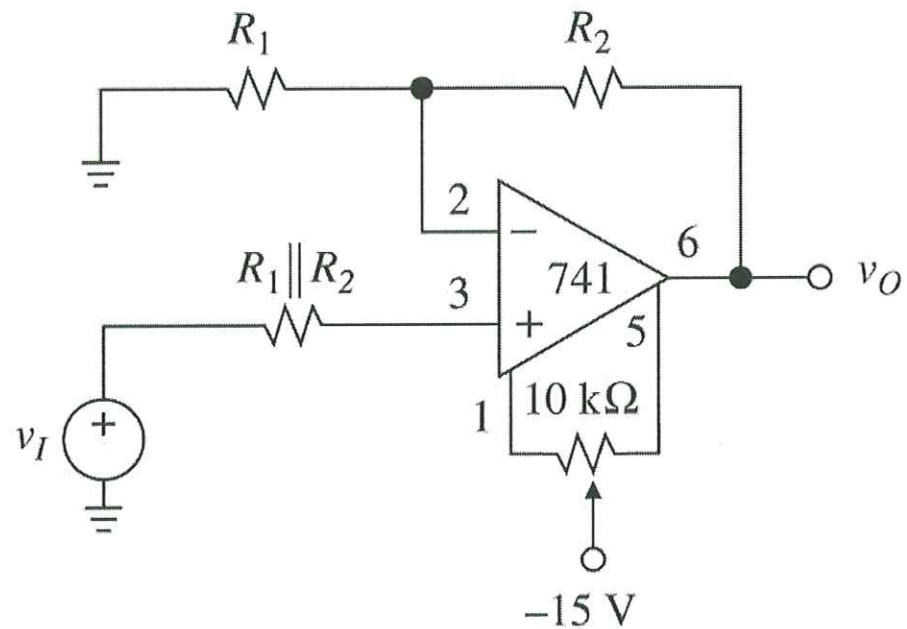


Figure 5.6: Detailed diagram of the input stage of the 741 op amp.

Application of previous



(a)



(b)

Figure 5.18: (a) Inverting, and (b) non-inverting amplifiers with internal offset nulling.

Same as previous,
different topology.

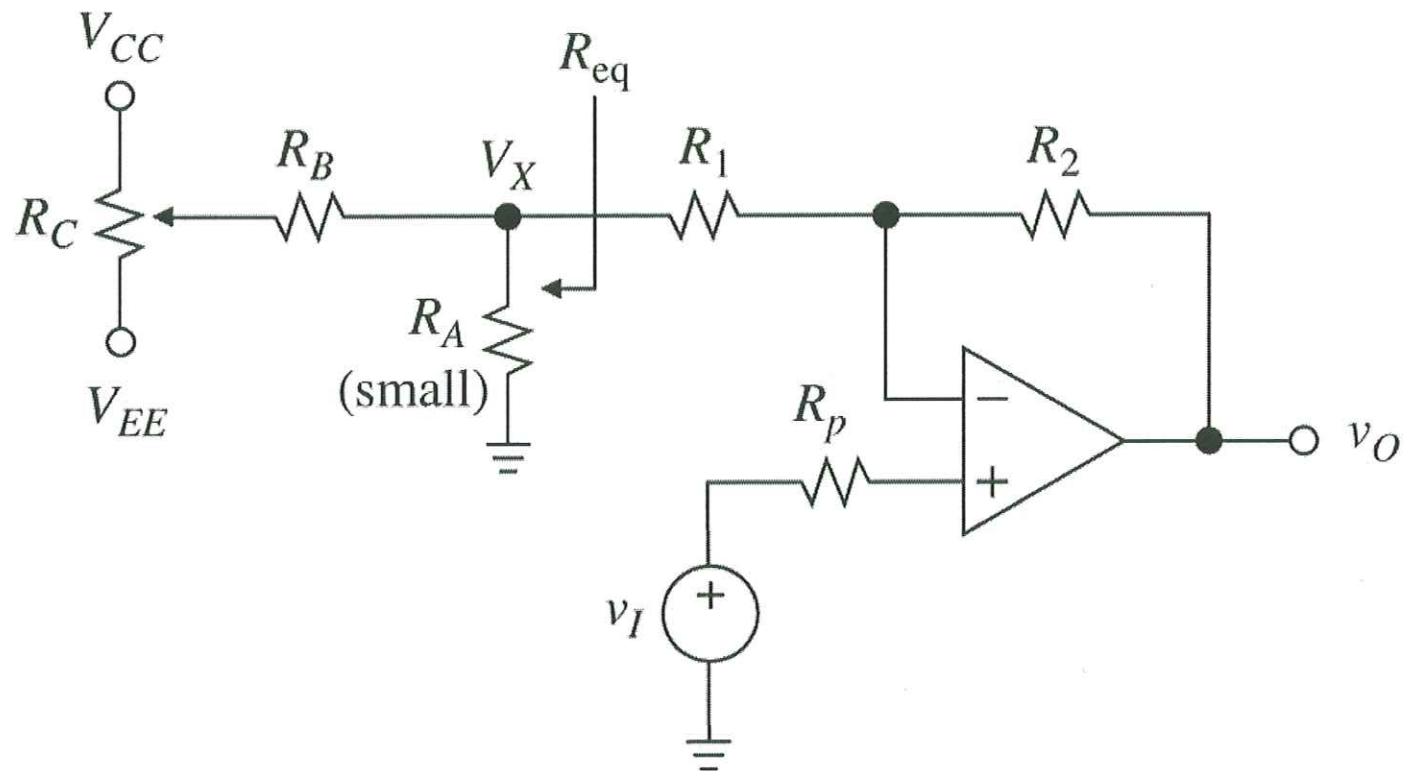
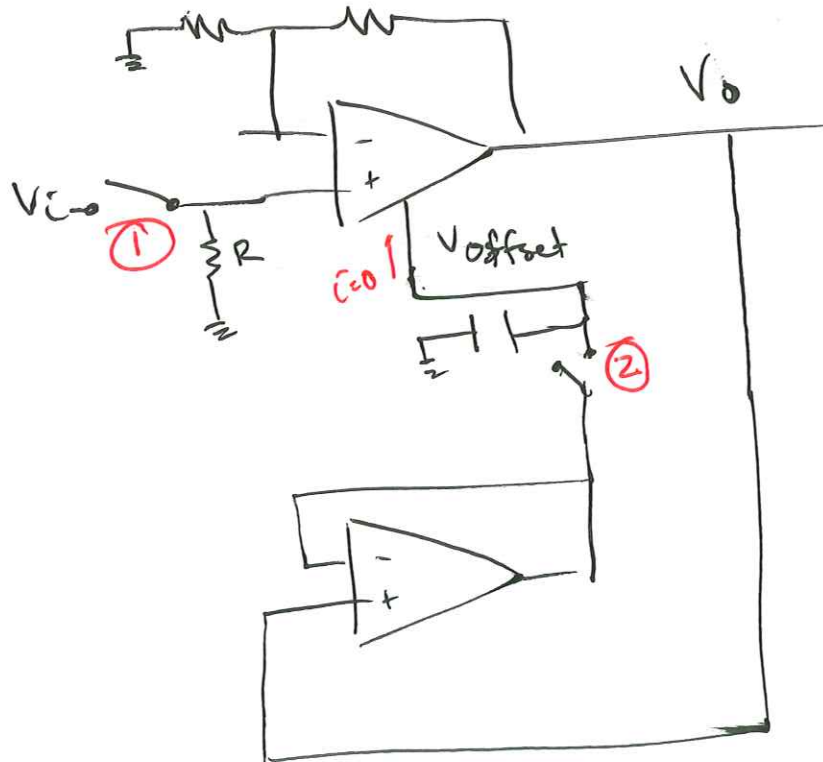


Figure 5.21: External offset error nulling for the non-inverting amplifier.

Chopper Circuits

(Probably drawn wrong)



Normal operation

sw ① closed

sw ② open

Calibration operation

sw ① open

sw ② closed

→ V_{offset} is a measure of output error

Toggling of switches would occur periodically to update V_{OE} measure at V_{offset} ...